

[58]

**SARDAR PATEL UNIVERSITY****Master of Science (Information Technology) (Semester-I) Examination - 2023****PS01CINT55: Operating System Concepts****Date: 21/04/2023 (Friday)****Time: 10:00 A.M. to 01:00 P.M.****Total: 70 Marks****NOTE:** Figure to the right indicates full marks of the questions.**Q.1. Multiple Choice Questions.****[08]**

1. One of the most important aspects of operating systems is the ability of \_\_\_\_\_.  
A. Swapping  
B. Output Friendly  
C. User Friendly  
D. Multitasking
2. \_\_\_\_\_ is a heart of OS and works as an interface between users and OS.  
A. Server  
B. Kernel  
C. User  
D. None of these
3. Full form of PCB is \_\_\_\_\_.  
A. Program Counter Block  
B. Process Control Block  
C. Portable Control Block  
D. Peterson's Algorithm
4. Run time mapping from virtual to physical address is done by \_\_\_\_\_.  
A. Memory management unit  
B. PCI  
C. CPU  
D. None of these
5. What a virtual-memory miss is called?  
A. Page miss  
B. Hit miss  
C. Page hit  
D. Page fault
6. Operating System maintains the page table for \_\_\_\_\_.  
A. Each Instruction  
B. Each Address  
C. Each Process  
D. Each Thread
7. Memory fragmentation results in \_\_\_\_\_.  
A. Stack overflow  
B. Page fault  
C. Poor utilization of memory  
D. Better utilization of memory
8. SAN stands for \_\_\_\_\_.  
A. Structure Area Network  
B. Structured Architecture Network  
C. Storage Area Network  
D. None of these

**Q.2. Answer the following questions (Attempt any Seven).****[14]**

1. What is an operating system?
2. Define an interrupt in an operating system.
3. Define response time.

**[PTO]**

4. List out various state of a process life cycle.
5. State two benefits of a multiprocessor system.
6. What is virtual memory?
7. What is file? List out different types of operations performed on a file.
8. What is FCFS?
9. What is demand paging?

**Q-3. Answer the following questions.**

- A. Explain various functions of an operating system in detail. [06]
- B. Discuss various types of Operating System (OS). [06]

**OR**

- B. Discuss about system calls of an operating system. [06]

**Q-4. Answer the following questions.**

- A. What is scheduling? Discuss the criteria affecting the scheduler's performance. [06]
- B. Explain Shortest Remaining Time (SRT) scheduling. [06]

**OR**

- B. Write the difference between Semaphore and Monitor. [06]

**Q-5. Answer the following questions.**

- A. Write the differences between paging and segmentation. [06]
- B. When does a page fault occur? Explain the Optimal Page Replacement Algorithm. [06]

**OR**

- B. Discuss about Contiguous Memory Allocation. [06]

**Q-6. Answer the following questions.**

- A. Discuss about SCAN and C-SCAN Disk Scheduling Algorithms. [06]
- B. Differentiate between NAS and SAN. [06]

**OR**

- B. Define RAID and discuss its various levels. [06]

\*\*\*\*\*X\*\*\*\*\*

(2)